

Exercises: Duality

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Exercise 1 (Dual formulation). Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$. iff $y \in x + K$. Consider the following program, Show that with $A \in M_{m,n}$ and $b \in \mathbb{R}^m$.

1. $\mathbb{I}_{g(x)=0} = \sup_{\lambda \in \mathbb{R}^m} \lambda^\top g(x)$
 2. $\mathbb{I}_{g(x) \leq 0} = \sup_{\lambda \in \mathbb{R}_+^m} \lambda^\top g(x)$
 3. $\mathbb{I}_{g(x) \in C} = \sup_{\lambda \in -C^\oplus} \lambda^\top g(x)$ where C is a closed convex cone, and $C^\oplus := \{\lambda \in \mathbb{R}^m \mid \lambda^\top c \geq 0, \forall c \in C\}$.
- (P) $\text{Min}_{x \in \mathbb{R}^n} c^\top x$
s.t. $Ax = b$
 $x \preceq_K 0$

Exercise 2 (Linear Programming). Consider the following linear problem (LP)

$$(P) \quad \text{Min}_{x \geq 0} c^\top x$$

s.t. $Ax = b$

1. Show that the dual of (P) is an LP.
2. Show that the dual of the dual of (P) is equivalent to (P).

Exercise 3 (Quadratically Constrained Quadratic Programming). Consider the problem

$$(QCQP) \quad \text{Min}_{x \in \mathbb{R}^n} \frac{1}{2} x^\top P_0 x + q_0^\top x + r_0$$

$$\frac{1}{2} x^\top P_i x + q_i^\top x + r_i \leq 0 \quad \forall i \in [m]$$

where $P_0 \in S_{++}^n$ and $P_i \in S_+^n$.

1. Show by duality that, for $\mu \in \mathbb{R}_+^m$, there exists P_μ, q_μ and r_μ , such that $g(\mu) = -\frac{1}{2} q_\mu^\top P_\mu^{-1} q_\mu + r_\mu \leq \text{val}(P)$.
2. Give an easy condition under which $\text{val}(P) = \sup_{\mu \geq 0} g(\mu)$.

Exercise 4 (Conic Programming). Let $K \subset \mathbb{R}^n$ be a closed convex pointed cone, and denote $x \preceq_K y$

1. Show that (P) is a convex optimization problem.

2. Denote $\mathcal{L}(x, \lambda, \mu) = c^\top x + \lambda^\top (Ax - b) + \mu^\top x$. Show that $\text{val}(P) = \text{Min}_{x \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}^m, \mu \in K^\oplus} \mathcal{L}(x, \lambda, \mu)$.

3. Give a dual problem to (P).

Exercise 5 (Duality gap). Consider the following problem

$$\text{Min}_{x \in \mathbb{R}, y \in \mathbb{R}_*^+} e^{-x}$$

s.t. $x^2/y \leq 0$

1. Find the optimal solution of this problem.
2. Write and solve the (Lagrangian) dual problem. Is there a duality gap ?

Exercise 6 (Two-way partitionning). Let $W \in S_n$ be a symmetric matrix, consider the following problem.

$$(P) \quad \text{Min}_{x \in \mathbb{R}^n} x^\top W x$$

s.t. $x_i^2 = 1 \quad \forall i \in [n]$

1. Consider a set of n element that you want to partition in 2 subsets, with a cost $c_{i,j}$ if i and j are in the same set, and a cost $-c_{i,j}$ if they are in a different set. Justify that it can be solved by solving (P).

2. Is (P) a convex problem ?
3. Show that, for any $\lambda \in \mathbb{R}^n$ such that $W + \text{diag}(\lambda) \succeq 0$, we have $\text{val}(P) \geq -\sum \lambda_i$. Deduce a lower bound on $\text{val}(P)$.

Exercise 7 (Linear SVM : duality (hard-margin)). Let $(x_i, y_i)_{i \in [n]}$ be labeled data with $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$. Consider the hard-margin SVM primal problem

$$(P) \quad \begin{aligned} & \underset{w \in \mathbb{R}^d, b \in \mathbb{R}}{\text{Min}} \quad \frac{1}{2} \|w\|_2^2 \\ & \text{s.t.} \quad y_i (w^\top x_i + b) \geq 1 \quad \forall i \in [n]. \end{aligned}$$

1. In which case can we guarantee strong duality for (P) ?
2. Derive the Lagrangian dual and express an optimal primal solution $(w^\sharp, b^\sharp)$ in terms of an optimal dual solution.

Exercise 8. We consider the following problem.

$$\underset{x_1, x_2}{\text{Min}} \quad x_1^2 + x_2^2 \quad (1)$$

$$\text{s.t.} \quad (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \quad (2)$$

$$(x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \quad (3)$$

1. Classify this problem (After 5th course)
2. Find the optimal solution and value of this problem.
3. Write and solve the KKT equation for this problem.
4. Derive and solve the Lagrangian dual of this problem.
5. Do we have strong duality ? If yes, could we have known it from the start ? If not, can you comment on why ?